

PLANT KINGDOM — REVISION NOTES

Key Concepts, Formulae, Example Mnemonics & Exam Highlights

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Target: NEET / Class 11 Last-Minute Revision

1. SYSTEMS OF CLASSIFICATION — QUICK SUMMARY

System	Proposed By	Key Criteria / Basis	Important Remarks
Artificial	Carolus Linnaeus	Superficial vegetative traits & androecium structure	Gave equal weight to vegetative & sexual traits. Rejected because vegetative traits change with environment.
Natural	Bentham & Hooker	Morphology, anatomy, embryology, phytochemistry	Considers external and internal features in detail.
Phylogenetic	Engler & Prantl	Evolutionary relationships	Assumes organisms in same taxa share a common ancestor.
Numerical	Computers / OTUs	All observable characteristics (codes/numbers assigned)	Hundreds of traits considered simultaneously; equal weight.
Cytotaxonomy	Cytologists	Chromosome number, structure, and meiotic behavior	Used to resolve complex taxonomic confusions.
Chemotaxonomy	Phytochemists	Chemical constituents (secondary metabolites, proteins)	Resolves ambiguities in traditional classification.

2. ALGAE — HIGH-YIELD FLASH CARDS

💡 Example Mnemonics for Algae Classes:

- **Chlorophyceae (Green):** "Very Cool Chlamydomonas Uses Spirogyra" → **V**olvox, **C**hara, **C**hlamydomonas, **U**lothrix, **S**pirogyra.
- **Paeophyceae (Brown):** "Please Ectocarpus Do Laminaria Sargassum Fucus" → **E**ctocarpus, **D**ictyota, **L**aminaria, **S**argassum, **F**ucus.
- **Rhodophyceae (Red):** "PG PG" → **P**olysiphonia, **G**racilaria, **P**orphyra, **G**elidium.

Class	Pigments	Stored Food	Cell Wall	Flagella
Chlorophyceae	Chl <i>a, b</i>	Starch (Pyrenoids) & Oil	Inner Cellulose, Outer Pectose	2–8, equal, apical
Phaeophyceae	Chl <i>a, c</i> , Fucoxanthin	Laminarin / Mannitol	Cellulose + Algin coating	2, unequal, lateral (pyriform)
Rhodophyceae	Chl <i>a, d</i> , r-Phycoerythrin	Floridian Starch	Cellulose, Pectin, Polysulphate esters	ABSENT (Non-motile)

Key Commercial Products of Algae:

- **Algin** Brown Algae (water-holding hydrocolloid)
- **Carrageen** Red Algae (water-holding hydrocolloid)
- **Agar** *Gelidium* & *Gracilaria* (microbe culture, ice creams, jellies)
- **Space Food** *Chlorella* & *Spirulina* (Unicellular protein-rich supplements)

3. BRYOPHYTES — QUICK POINTS

- Known as "**Amphibians of the Plant Kingdom**" because they require water for fertilization (motile antherozoid transport).
- Dominant phase: Haploid **Gametophyte (n)**. Sporophyte (2n) is dependent on gametophyte for nutrition.
- Root-like structures: **Unicellular rhizoids** (Liverworts) or **Multicellular/branched rhizoids** (Mosses).
- **Liverworts (e.g., *Marchantia*)**: Asexual reproduction by **Gemmae** (green, multicellular asexual buds inside Gemma Cups).
- **Mosses (e.g., *Funaria*, *Sphagnum*)**: Gametophyte has two stages:
 1. *Protonema stage*: Creeping, green, filamentous (from spore).
 2. *Leafy stage*: Erect, spirally arranged leaves, bears sex organs (from secondary protonema).
- **Economic Use**: *Sphagnum* produces **Peat** (fuel & packing material for living trans-shipment due to water retention).

4. PTERIDOPHYTES — VASCULAR PIONEERS

Milestone: First terrestrial plants with **vascular tissue (Xylem & Phloem)**. Dominant phase is **Sporophyte (2n)**.

- **Leaf Types**: *Microphylls* (small, e.g., *Selaginella*) vs *Macrophylls* (large, e.g., Ferns).
- **Sporophylls**: Leaf-like structures bearing sporangia; form compact **Strobili/Cones** in *Selaginella* and *Equisetum*.
- **Prothallus**: Inconspicuous, free-living, photosynthetic thalloid gametophyte (n). Requires cool, damp places & water for fertilization.
- **Heterospory & Seed Habit**:
 - *Homosporous*: Most pteridophytes (spores identical).
 - *Heterosporous*: *Selaginella* and *Salvinia* (produces Macro/Megaspores and Microspores).
 - Female gametophyte retained on parent sporophyte → **Precursor to Seed Habit**.

Class	Key Examples
Psilopsida	<i>Psilotum</i>
Lycopsidea	<i>Selaginella</i> , <i>Lycopodium</i>
Sphenopsida	<i>Equisetum</i>
Pteropsida	<i>Dryopteris</i> , <i>Pteris</i> , <i>Adiantum</i> , <i>Salvinia</i> , <i>Marsilea</i>

5. GYMNOSPERMS — NAKED SEEDS

- **Naked Seeds**: Ovules are NOT enclosed by ovary wall before or after fertilization.
- **Root Associations**:
 - **Mycorrhiza**: Fungal association in *Pinus* roots.
 - **Coralloid Roots**: Associated with N_2 -fixing cyanobacteria in *Cycas*.
- **Xerophytic Adaptations**: Needle-like leaves, thick cuticle, and **sunken stomata** (reduces transpiration).
- **Gametophyte Reduction**: Gametophytes (n) have no independent existence; completely retained within sporangia on parent sporophyte. Male gametophyte is reduced to a **Pollen Grain**.

Microspore



Pollen Grain



Wind Transfer



Pollen Tube to Archegonia



Naked Seed

6. COMPARATIVE MASTERY TABLE

Feature	Algae	Bryophytes	Pteridophytes	Gymnosperms	Angiosperms
Dominant Phase	Gametophyte (n)	Gametophyte (n)	Sporophyte (2n)	Sporophyte (2n)	Sporophyte (2n)
Vascular Tissue	Absent	Absent	Present	Present	Present
Main Body	Thalloid	Thalloid / Leafy	Root, Stem, Leaves	Root, Stem, Leaves	Root, Stem, Leaves
Fertilization Medium	Water	Water	Water	Wind / Pollen Tube	Pollination Vectors
Seed & Fruit	None	None	None (Precursor)	Naked Seed (No Fruit)	Enclosed Seed in Fruit